The replicate and variance projection

How the calculated significance changes when an assumed replicate count or variance is substituted while the observed effects are held fixed.

AT A GLANCE

Where	Statistics → Replicate & Variance Projection, titled Projected volcano plot
What it is	A what-if projection, not a power calculation.
Data levels	Protein, phosphosite, phosphoprotein
Needs	The same two conditions a volcano needs, each with at least two samples
What changes	The replicate count and the variance that go into the standard error. Nothing else.
What does not	The fold changes. They are the measured ones, and no setting on this page moves a point sideways.
Not random	Nothing is resampled, drawn or generated. There is no seed, and the same settings always give the same figure.
Renderer	Interactive only. There is no static figure and no appearance panel on this tab.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

1 What it simulates

The controls under **Projection settings** have the following interpretation:

> **INTERPRETING THE PROJECTION**
>
> A what-if projection, not a power calculation: the observed mean fold changes are held fixed and the t-test is recomputed for the variance factor and replicates per group set here. Nothing is drawn at random, so the same settings always give the same plot. The observed fold changes include noise, so the projection is optimistic for effects that look large, and it does not estimate the probability of detecting a true effect.

The recomputation is arithmetic, not an experiment. For every feature the observed condition means and the observed within-condition variances are taken from the selected matrix, exactly as the volcano takes them. The variances are then multiplied by **Variance × factor**, the replicate counts are replaced by **Replicates per group** if a fixed count is set, and the same two-sample t-test is recomputed with those substituted numbers. The fold change is untouched, so every point stays where the real volcano put it; only its height changes.

That is the whole of it. No values are drawn at random, no dataset is generated, and no feature is invented: the page is a what-if on the *precision* of the effects already in the data.

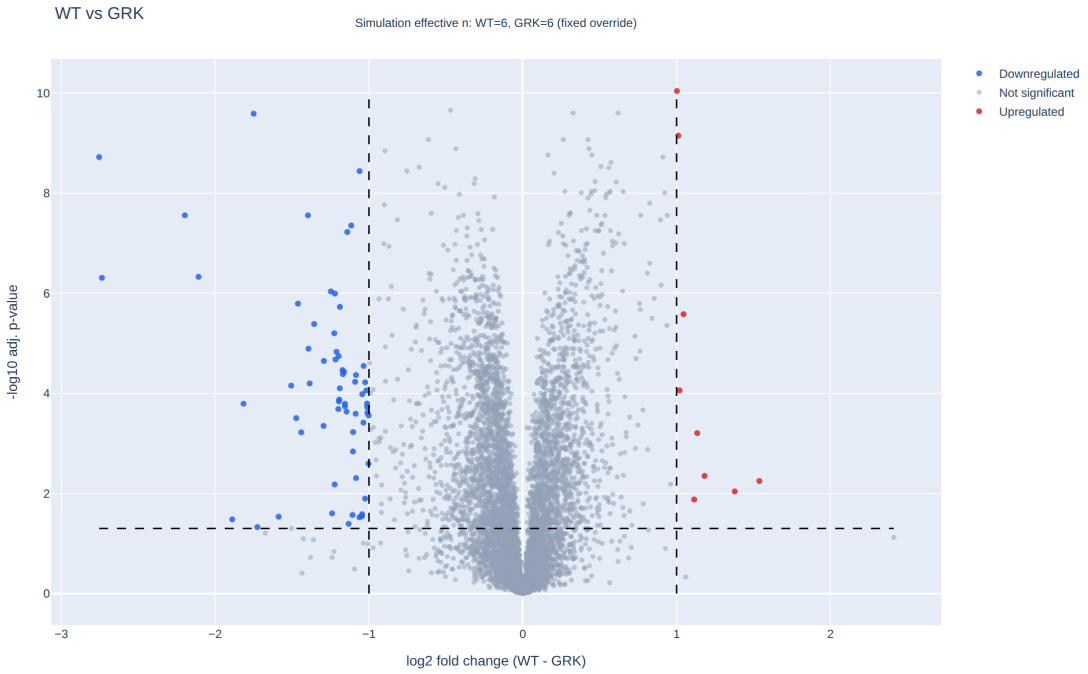


Figure 1.1. The projected volcano for WT against GRK at protein level, with the replicate count fixed at 6. The line above the plot states the effective *n* the simulation used and where it came from; the axes are the measured $\log_2$ fold change and the p-value the test would return under that substituted count. The three dashed lines are the two fold change thresholds and the p-value threshold.

Reading it

- The line above the plot: the effective *n* actually used for each condition, as a single number or a range, and whether it came from the data (observed) or from the control (fixed override).
- The x axis: the measured $\log_2$ fold change, Condition 1 − Condition 2, identical to the volcano page.

- The y axis: $-\log_{10}$ of the p-value the test *would* return under the substitution. This is the only thing the two simulation controls move.
- The colours and the three threshold lines, read exactly as on the volcano.
- The summary above the plot: the method, the sample-size mode, the effective n of each condition, the number of rows, and how many features are called up, down, not significant, or have no p-value under these settings.

Holding an observed effect fixed while raising the assumed replicate count lowers its standard error. The figure shows the arithmetic consequence of that assumption and nothing more: a candidate crossing the line at a fixed count of six is significant in the substituted calculation under those fixed inputs, not a prediction of what an experiment with six replicates would find.

> **EASY TO GET WRONG**
>
> A what-if projection, not a power calculation. Every feature keeps its own measured effect and its own measured variance, so a variance estimated from three replicates is no more reliable at a fixed count of 20 than it was at three; the page says nothing about effects this dataset did not detect at all. The observed fold changes include noise, so the projection is optimistic for effects that look large, and it does not estimate the probability of detecting a true effect.

2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Protein, phosphosite or phosphoprotein.	Protein	page
Source	Fixed at Volcano: the projection always starts from a two-condition comparison.	Volcano	page
Data Source	Which matrix the observed means and variances come from.	the most processed matrix available	page
Condition 1, Condition 2	The comparison. Condition 1 is the minuend, as on the volcano page.	first condition in the annotation	page
Label column	Which column names a feature in the hover text: the workflow name or the gene names.	the workflow name	page

Controls on this page (continued)

Control	What it does	Default	Where
Test type	Unpaired (Student's t, equal variance) or Unpaired (Welch, unequal variance). The paired test is not offered here.	Unpaired (Student's t, equal variance)	page
P-value threshold, log2 FC threshold	The two thresholds the simulated result is classified against.	0.05, 1	page
Use uncorrected p-values	Classifies on the raw simulated p-value instead of the adjusted one.	off	page
Variance × factor	Multiplies both within-condition variances before the standard error is formed. Below 1 is a cleaner experiment, above 1 a noisier one; the field clamps to a minimum of 0.01.	1	page
Replicates per group (0 = observed, 2-100 fixed)	0 keeps each feature's own observed replicate counts; a value from 2 to 100 replaces both counts with that number for every feature. 1 is refused, because a t-test cannot estimate variance from one sample per condition.	0	page

WORTH KNOWING

This page has no plot appearance panel: the figure it draws is not one of the styled plots, so **Ctrl** + **Shift** + **E** adds nothing here.

3 How it is computed

THE SUBSTITUTION

A feature needs at least two observed values in each condition, as on the volcano; everything else is dropped before anything is computed. For a surviving feature, with $\bar{x}, \bar{y}$ the observed condition means, s_1^2, s_2^2 the observed within-condition variances and $n_1^{\text{obs}}, n_2^{\text{obs}}$ the observed counts:

$$\log_2 \text{FC} \;=\; \bar{x} - \bar{y} \qquad \text{(unchanged by any setting on this page),}$$

$$s_k'^2 = m \cdot s_k^2, \qquad n_k' = \begin{cases} n_k^{\text{obs}} & \text{override} = 0, \\ n^{\text{fixed}} & \text{otherwise,} \end{cases}$$

where m is **Variance** × **factor** and n^{fixed} is **Replicates per group**. Note that n_k' replaces the count in the standard error and in the degrees of freedom only; the variance it is divided by is still the one estimated from the replicates that were actually measured.

THE TEST

With the substituted numbers, the pooled option forms

$$s_p'^2 = \frac{(n_1' - 1)s_1'^2 + (n_2' - 1)s_2'^2}{n_1' + n_2' - 2}, \quad SE = \sqrt{s_p'^2 \left(\frac{1}{n_1'} + \frac{1}{n_2'}\right)}, \quad df = n_1' + n_2' - 2,$$

and the Welch option forms $SE = \sqrt{s_1'^2/n_1' + s_2'^2/n_2'}$ with Welch–Satterthwaite degrees of freedom. Either way

$$t = \frac{\log_2 \text{FC}}{SE}, \qquad p = 2\,P(T_{df} > |t|).$$

Because SE falls as $1/\sqrt{n'}$ while $\log_2$ FC is fixed, raising the replicate count moves every point straight up the figure and never sideways.

CORRECTION AND CLASSIFICATION

The simulated p-values are adjusted by Benjamini–Hochberg over the features tested, unless **Use uncorrected p-values** is on, and classified against the two thresholds exactly as on the volcano page: Upregulated above $+\tau$ and below α, Downregulated below $-\tau$ and below α, Unavailable where no finite p-value resulted, Not significant otherwise. The correction is therefore run on a set of p-values that were never observed, which is the point — but it means the adjusted values are as hypothetical as the raw ones.

Edge cases

- An override applies the same n to every feature, including one observed in only two of five samples. The summary reports both the mode and the effective n so the substitution is never silent.
- With the override at 0, the effective n is a range rather than a number whenever coverage differs between features, and the annotation above the plot shows it as one.
- Zero variance in both conditions gives zero standard error. Distinct constant means produce an extreme test statistic; identical constant means give an undefined test, not evidence

of a difference. Scaling zero cannot supply missing variability.

- A saved panel state holding the old value 1, or any other value the field cannot hold, falls back to 0 on load, because a t-test cannot estimate variance from one sample per condition. Typing 1 is refused with a message.
- This tab is not available in viewer mode.

4 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
fig = volcano_plot_sim(data, meta, "WT", "GRK", workflow="Protein", mod_var=1,
    ↪ mod_n=6)
```

R

```
fig <- volcano_plot_sim(data, meta, "WT", "GRK", workflow = "Protein", mod_var = 1,
    ↪ mod_n = 6)
```

This is one call rather than the page's two stages: mod_var and mod_n are **Variance × factor** and **Replicates per group**, and mod_n=0 keeps each feature's own observed counts exactly as 0 does on the page. Two things the page offers are not here — there is no Welch option, because the function forms the pooled variance and nothing else, and there is no summary line, so the effective *n* has to be read from the override you set; what comes back is the Plotly figure, which is the only renderer either side has for this plot.

5 Exporting

There is one renderer and no download button: Open Plot opens the interactive figure in its own tab, where the browser's own save is the only export. There is no static version of this plot and no result table under it, so a simulated comparison that has to appear in a document is best described by its settings — the two conditions, the test, the variance factor and the effective *n*; the summary above the plot states the test and the effective *n*.

The page reports counts and a figure, not a per-feature export. Do not report these hypothetical p-values as experimental findings: the page is a what-if projection, not a power calculation,

and it does not estimate the probability of detecting a true effect.

Behind this page

Function	What it produces
volcano_plot_sim	The projected volcano. Its mod_var and mod_n arguments are **Variance × factor** and **Replicates per group**; it offers the pooled variance model only.